Mlytics: Empowering digital businesses with a powerful experience delivery platform on Google Cloud

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mlytics

About Mlytics

Mlytics is a Singapore-based experience delivery platform. It empowers digital businesses by deploying AI algorithms to automatically direct website end users to the fastest CDN available. Partnering with the world's leading CDN providers, Mlytics aggregates CDN performance through a proprietary decision engine that enables near zero-latency connections around the clock. With offices in Singapore, Taiwan, Hong Kong, and the UK, Mlytics has expanded from Asia into Europe and now is developing a global growth strategy.

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Industries: Technology

Location: Hong Kong

(https://cloud.google.com/memorystore/),
BigQuery (https://cloud.google.com/bigquery/),

Google Cloud (https://cloud.google.com/)

Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)

Cloud Memorystore (https://cloud.google.com/memorystore/)

BigQuery (https://cloud.google.com/bigquery/)

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Compute Engine (https://cloud.google.com/compute/)

Google Cloud results

- Enables AI decision engines by auto scaling computing power when modeling big data points
- Boosts creative mission by freeing up 20% in fixed costs for investment in solution development
- Enables global expansion strategy through listing on Google

Mana
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Cloud Marketplace and Google ecosystem integration
- Prevents failure around the clock with replica nodes on Memorystore

From ecommerce sites to gaming platforms and livestreaming apps, today's digital businesses rely increasingly on content delivery networks (CDNs) to thrive in competitive markets.

No matter how fabulous the app, slow load times and service outages can lead to significant lost business in an age where consumers expect near zero-latency service around the clock. CDNs solve this problem by deploying a network of servers around the globe to deliver content as fast as possible to the end user.

However, even the most powerful CDNs are prone to outages. And at any given time, due to factors ranging from traffic volume to network stability, some CDNs will be faster than others. That's what gave a group of digital innovators based out of Taipei a powerful idea. What if it could use AI-powered data analytics to aggregate the performance of multiple CDNs and automatically link customers to the fastest one at any point in time?

"Building a successful multi-CDN solution that spans the globe and always enables the fastest connections requires the most powerful cloud architectures and tools. Google Cloud provides the unlimited autoscaling, security, and versatility we need to deliver on our commitments to digital enterprises."

—Locarno Pan, Managing Partner, Mlytics

The brainwave led to the launch of Mlytics (https://www.mlytics.com/), a growing operation that partners with the world's leading CDN providers to provide digital businesses and end users with fast and stable connectivity around the clock. The solution works by monitoring millions of data points from both partner CDNs and client websites, as well as performing simulated "synthetic" data monitoring and running these datastreams through AI decision engines. The system then continuously switches customers back and forth between the best-performing CDNs based on the user’s geolocation.

To succeed, Mlytics needed to run its data pipelines and decision engines on the most powerful autoscaling cloud infrastructure available, in order to handle the massive data traffic fluctuations emanating from customers and CDN partners. It also sought top-flight machine learning (ML) tools for the next phase of its evolution, which is to build decision engines able to predict, rather than react to, CDN performance. Mlytics turned to a Google Cloud (https://cloud.google.com/) and found the unlimited scale and ML data-processing power it needed to develop the future of AI-driven multi-CDN solutions.

"Building a successful multi-CDN solution that spans the globe and always enables the fastest connections requires powerful cloud architectures and tools," says Locarno Pan, Managing Partner at Mlytics. "Google Cloud provides the unlimited autoscaling, security, and versatility we need to deliver on our commitments to digital enterprises."

"The autoscaling enabled by Google Cloud is really what enables our CDN decision engines to make instant decisions based on data and traffic flows. When there's a huge spike in user demand coming in, we know that with Compute Engine and Google Kubernetes

Engine we can absorb it easily."

—Edward Hu, Vice-President of Product & Growth, Mlytics

Connecting businesses to speed and scale with Compute Engine and GKE

Mlytics needs to host an impressive array of complex and computing power-heavy assets on its cloud infrastructure. In addition to its AI-based decision engines, these include a stream-processing solution for real-time data feeds and proprietary Web Application Firewall to keep its client and partner CDN data secure.

Google Cloud was the natural choice for building the Mlytics platform because its engineers loved its ease of use. The critical factor, however, was how Google Cloud infrastructure provided the startup the power and versatility it needed to succeed.

In particular, the autoscaling virtual machine capabilities of Compute Engine (/compute) enable Mlytics to handle the massive data traffic churning through its engines every second, even during times of heavy traffic, which can be 200% more than the usual traffic load.

Meanwhile, the seamless and granular containerized scalability of Google Kubernetes Engine (/kubernetes-engine) (GKE), allows the Mlytics decision engines to automatically boost computing power when modeling big data points. This has helped save DevOps time by almost 100% as they don't have to worry about troubleshooting and can focus on building better features as the business grows.

"The autoscaling enabled by Google Cloud is really what enables our CDN decision engines to make instant decisions based on data and traffic flows," says Edward Hu, Vice-President of Product and Growth at Mlytics. "When there's a huge spike in user demand coming in, we know that with Compute Engine and Google Kubernetes Engine we can absorb it easily."

Mlytics also uses Memorystore (/memorystore) to help its DevOps team improve and ensure a Redis high availability. This is done by replicating a primary Redis node to a replica node, so that if a primary node fails, Google Cloud can trigger a failover and promote the replica to be the new primary node. This serves as a security layer, preventing a single point failure for customers at any time.

Coupled with the powerful abilities of these tools, the flexibility of Google Cloud cost modeling has been a major factor in enabling Mlytics to expand and evolve its solution. This has helped the company save 20% of its fixed costs, which can be invested in creating innovative platform enhancements.

Expanding global market reach with Google Cloud Marketplace

Mlytics has been able to translate extreme CDN accuracy, security, and reliability into fast expansion across Asia and into Europe. Now it is planning to spread its solution around the world.

As a startup, however, Mlytics cannot afford to hire a sales team to help it scale globally. That's why Mlytics is working with Google Cloud to get listed on Google Cloud Marketplace (/marketplace), a network of Google Cloud-approved independent software providers (ISPs) meeting enterprise IT needs within the Google ecosystem.

Since Mlytics runs its core solutions and backend systems on Google Cloud, Pan says it makes sense to leverage Google's massive global reach through Marketplace as the main enabler of its global expansion.

"By listing on Google Cloud Marketplace, we're looking forward to deepening our partnership with Google Cloud, and by doing so win new partner clients around the world," says Pan. "Driving our marketing through our principal cloud provider will enable us to scale our business globally."

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—Locarno Pan, Managing Partner, Mlytics

Building the future of CDN with machine learning driven by BigQuery

The next step in the Mlytics journey is to train its decision engines to carry out performance predictions by using machine learning for big data analysis of CDN dataflows. Currently, Mlytics crunches the data in a more reactive mode, identifying issues in the network and fixing them instantly, by switching clients to the optimal provider.

Transitioning to a predictive model will enable Mlytics to offer an even faster and safer CDN experience by anticipating traffic trends, network stability conditions, and potential security threats. The startup has begun experimenting with BigQuery (/bigquery) to enable the ML big data analytics necessary to turn this vision into reality.

"With BigQuery, we'll be able to predict CDN performance slippages, traffic bottlenecks, or security threats, so we can be more proactive supporting our customers," says Pan. "We're excited by the unlimited potential of Google Cloud to enable us to continuously evolve our solution and build the future of multi CDN."